Simultaneous Equations & Inequalities

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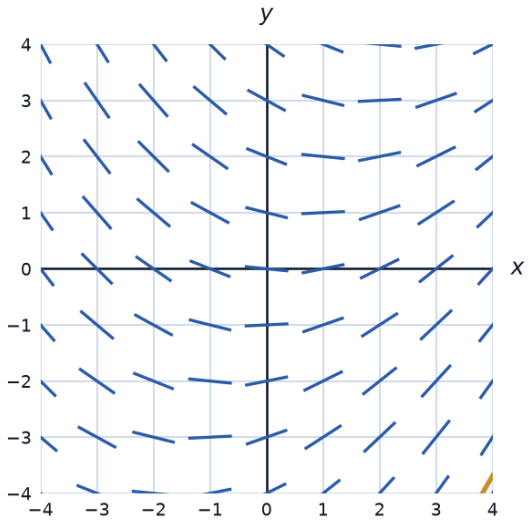
Part I — Free-Response Questions (FRQs): Foundational Fluency (1–7)

1 [Foundational] • Target: Free-Response Questions (FRQs): Foundational Fluency (1–7)

The temperature θ of a cooling metal rod satisfies $\frac{d\theta}{dt} = -k(\theta - 20)$, where t is time in minutes and k is a constant. Given $\theta = 80$ when $t = 0$ and $\theta = 50$ when $t = 10$, find the value of t when $\theta = 30$.

2 [Foundational] • Target: Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14)

[Differential Equations · Practice 3] If the governing relation for this system is $g(t) = 5t + 5$, evaluate $g(8)$ and find the inverse input t when $g(t) = 50$.



3 [Foundational] • Target: Free-Response Questions (FRQs): Challenging & Cross-Lesson Synthesis (15–20)

Solve the separable DE $\frac{dy}{dx} = \frac{x+1}{y}$ given $y = 2$ when $x = 0$.

4 [Foundational] • Target: Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)

Prove by mathematical induction that for all integers $n \geq 1$: $1 + 3 + 5 + \ \ldots\ + (2n - 1) = n^2$.

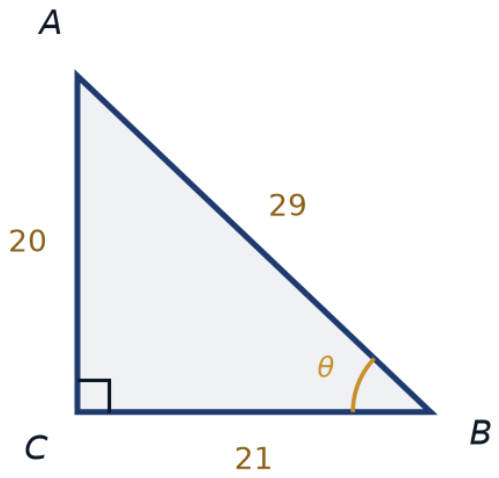
5 [Foundational] • Target: Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37)

Solve the trigonometric equation $\cos(2\theta) + 3\sin\theta = 2$ for all solutions $\theta \in [0, 2\pi)$.

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6 [Foundational] • Target: Multiple-Choice Questions (MCQs): Challenging & Cross-Lesson Synthesis (38–45)

In a 2D Cartesian plane, points A, B, C have coordinates $A(2, -1)$, $B(5, 3)$, and $C(-1, k)$. (a) Find the vectors $\vec{AB}$ and $\vec{AC}$ in component form. (b) Find the value of k for which $\vec{AB}$ is perpendicular to $\vec{AC}$. (c) For $k = 3$, calculate the exact cosine of the angle between $\vec{AB}$ and $\vec{AC}$.



7 [Foundational] • Target: Free-Response Questions (FRQs): Foundational Fluency (1–7)

Solve the equation $2\cos 2x = 1 + \sin x$ for $0 \le x \le 2\pi$ radians, giving answers to 3 significant figures.

Part II — Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14)

8 [Intermediate] • Target: Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14)

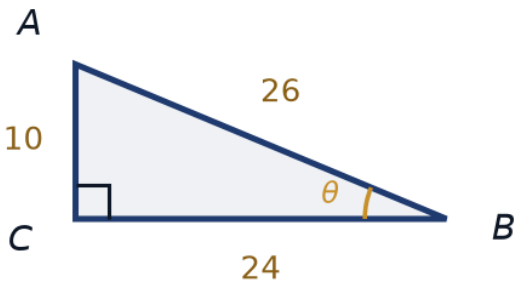
Consider the binomial expansion of $\left(x^2 - \frac{2}{x}\right)^9$. (a) Write down the general term in terms of r. (b) Find the coefficient of x^3. (c) Find the term independent of x.

9 [Intermediate] • Target: Free-Response Questions (FRQs): Challenging & Cross-Lesson Synthesis (15–20)

Find the constant term (independent of x) in the binomial expansion of $\left(2x^2 - \frac{1}{x}\right)^9$.

10 [Intermediate] • Target: Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)

The curve C has equation $x^3 - 2xy + y^2 = 4$. Find the coordinates of the two points on C where the tangent is parallel to the y-axis (vertical).



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11 [Intermediate] • Target: Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37)

Determine the slant asymptote of $g(x) = \frac{2x^3 - 3x^2 + 5}{x^2 + 1}$ and find the coordinates of any point where the curve intersects this asymptote.

12 [Intermediate] • Target: Multiple-Choice Questions (MCQs): Challenging & Cross-Lesson Synthesis (38–45)

Practice 3.1: Solve the equation $2\cos 2x = 4\cos x - 3$ for $0 \le x \le 2\pi$, giving exact solutions.

13 [Intermediate] • Target: Free-Response Questions (FRQs): Foundational Fluency (1–7)

Convert the Cartesian equation $(x^2 + y^2)^2 = 4(x^2 - y^2)$ into polar form and identify the name and key features of the resulting curve.

14 [Intermediate] • Target: Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14)

Find all vertical and horizontal asymptotes, the exact coordinates of any holes, and all intercepts of the rational function $R(x) = \frac{2x^2 + 5x - 3}{x^2 + 4x + 3}$.

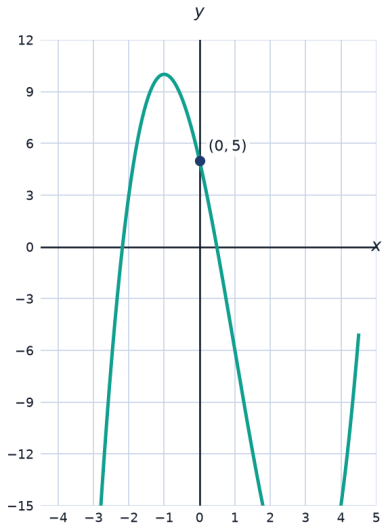
Part III — Free-Response Questions (FRQs): Challenging & Cross-Lesson Synthesis (15–20)

15 [Challenging] • Target: Free-Response Questions (FRQs): Challenging & Cross-Lesson Synthesis (15–20) • Cross-Lesson: 16.01 Permutations & Combinations

Solve the logarithmic equation $\log_3(x + 6) + \log_3(x) = 3$ and eliminate any extraneous solutions.

16 [Challenging] • Target: Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28) • Cross-Lesson: 17.01 Discrete Random Variables

Practice 5.1: A curve has equation $y = x^3 - 3x^2 - 9x + 5$. (a) Find the coordinates of the local maximum and local minimum points. (b) Determine the coordinates of the point of inflection and find the equation of the tangent at the point of inflection.



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17 [Challenging] • Target: Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37) • Cross-Lesson: 18.01 Binomial Distribution

Write the standard equation of the ellipse having foci at $(\pm 3, 0)$ and vertices at $(\pm 5, 0)$. State the lengths of the major and minor axes, and compute the eccentricity.

18 [Challenging] • Target: Multiple-Choice Questions (MCQs): Challenging & Cross-Lesson Synthesis (38–45) • Cross-Lesson: 19.01 Normal Distribution

Practice AI.2: A company takes out a loan of 120,000 to purchase equipment at an annual interest rate of 6% compounded monthly, repayable by equal monthly payments over 10 years. (a) Find the amount of each monthly payment. (b) Calculate the total interest paid over the life of the loan.

19 [Challenging] • Target: Free-Response Questions (FRQs): Foundational Fluency (1–7) • Cross-Lesson: 3.01 Graphs & Transformations

The first term of a geometric sequence is $u_1 = 18$ and the common ratio is $r = \frac{1}{3}$. (a) Find the exact value of the 5th term u_5. (b) Calculate the sum of the first 6 terms, S_6. (c) State why the sum to infinity S_∞ exists, and find its exact value.

20 [Challenging] • Target: Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14) • Cross-Lesson: 4.01 Coordinate Geometry

Practice 2.1: The function $f(x) = 3 - \ln(2x + 1)$ is defined for $x > -0.5$. Find an expression for $f^{-1}(x)$, state its domain and range, and write down the coordinates of the intercept of $y = f^{-1}(x)$ with the y-axis.

Part IV — Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)

21 [Foundational] • Target: Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)

Solve the simultaneous equations $y = 2x - 3$ and $x^2 + y^2 = 26$. Which pair of points represents the solutions?

a. $(1, -1)$ and $\left(\frac{17}{5}, \frac{19}{5}\right)$

b. $(3, 3)$ and $(-1, -5)$

c. $(5, 7)$ and $(1, -1)$

d. $(2, 1)$ and $(-2, -7)$

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22 [Foundational] • Target: Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)

The line $x + y = 5$ intersects the hyperbola $xy = 6$. Find the coordinates of the two intersection points.

a. $(2, 3)$ and $(3, 2)$

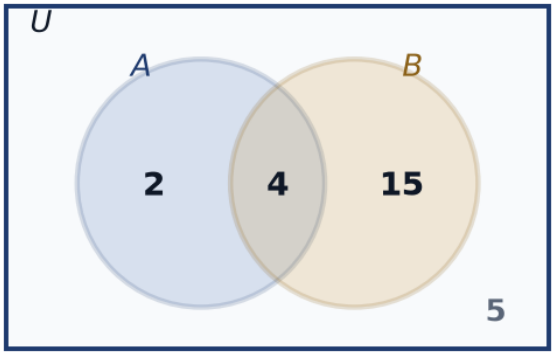
b. $(1, 6)$ and $(6, 1)$

c. $(2, 3)$ and $(-2, -3)$

d. $(4, 1)$ and $(1, 4)$

23 [Foundational] • Target: Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)

Find the points of intersection of the line $2x - y = 1$ and the curve $y = x^2 - 4$.



a. $(3, 5)$ and $(-1, -3)$

b. $(2, 3)$ and $(-2, -5)$

c. $(4, 7)$ and $(1, 1)$

d. $(5, 9)$ and $(0, -1)$

24 [Foundational] • Target: Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)

Solve the quadratic inequality $x^2 - 2x - 8 < 0$.

a. $-2 < x < 4$

b. $x < -2$ or $x > 4$

c. $-4 < x < 2$

d. $x < -4$ or $x > 2$

25 [Foundational] • Target: Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)

Find the set of values of x satisfying $2x^2 + 5x - 3 \geq 0$.

a. $x \leq -3$ or $x \geq \frac{1}{2}$

b. $-3 \leq x \leq \frac{1}{2}$

c. $x \leq -\frac{1}{2}$ or $x \geq 3$

d. $-1 \leq x \leq 3$

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26 [Foundational] • Target: Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)

Solve the simultaneous inequalities $3x - 5 < 7$ and $x^2 - 9 \geq 0$.

a. $x \leq -3$ or $3 \leq x < 4$

b. $-3 \leq x \leq 3$

c. $x < 4$

d. $3 \leq x < 4$

27 [Foundational] • Target: Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)

For what range of values of m does the line $y = mx$ not intersect the curve $y = x^2 + 9$?

a. $-6 < m < 6$

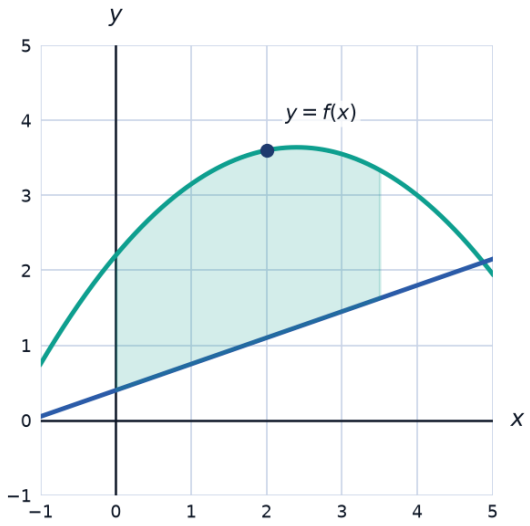
b. $m < -6$ or $m > 6$

c. $-3 < m < 3$

d. $m > 6$

28 [Foundational] • Target: Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)

The perimeter of a rectangle is 28 cm and its area is at least 45 cm^2. What is the range of possible values for the width w ($w \leq l$)?



a. $5 \leq w \leq 7$

b. $4 \leq w \leq 9$

c. $3 \leq w \leq 7$

d. $w \geq 5$

Part V — Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37)

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29 [Intermediate] • Target: Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37)

Find the coordinates of the points where the line $y = 3x - 2$ is normal to the curve $y = x^2 - 4x + 3$.

a. $(1, 0)$

b. $(2, -1)$

c. $(0, -2)$

d. $(3, 0)$

30 [Intermediate] • Target: Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37)

Find the range of values of k for which the line $y = kx - 1$ intersects the circle $x^2 + y^2 = 4$ in two distinct points.

a. $k^2 > -\frac{3}{4}$ (all real k)

b. $k > \sqrt{3}$

c. $k < -\sqrt{3}$ or $k > \sqrt{3}$

d. $-1 < k < 1$

31 [Intermediate] • Target: Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37)

A company finds that its profit P (in thousands of dollars) satisfies $P(x) = -2x^2 + 28x - 80 \geq 10$, where x is the production level in hundreds of units. What is the valid production interval for x?

a. $5 \leq x \leq 9$

b. $4 \leq x \leq 10$

c. $3 \leq x \leq 11$

d. $6 \leq x \leq 8$

32 [Intermediate] • Target: Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37)

Find all values of x satisfying the rational inequality $\frac{x-3}{x+2} < 0$.

a. $-2 < x < 3$

b. $x < -2$ or $x > 3$

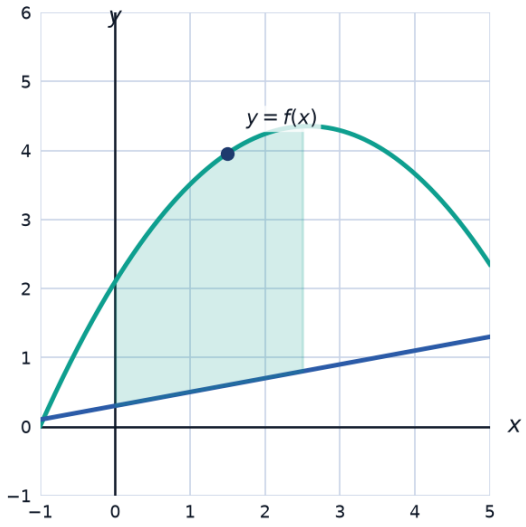
c. $x < 3$

d. $-3 < x < 2$

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33 [Intermediate] • Target: Free-Response Questions (FRQs): Foundational Fluency (1–7)

Relative to a fixed origin O, the lines L_1 and L_2 have vector equations $L_1 : \mathbf{r} = (1; 0; -1) + \lambda(1; 2; 2)$, $L_2 : \mathbf{r} = (2; -4; 7) + \mu(2; 1; -2)$. The unique points P on L_1 and Q on L_2 are such that the line segment PQ is perpendicular to both L_1 and L_2. A point R lies on L_1 such that the area of triangle PQR is 18. Given that the z-coordinate of R is positive, what is the exact distance from the origin O to the midpoint of the line segment QR?



a. $3\sqrt{5}$

b. $\sqrt{13}$

c. $6\sqrt{2}$

d. $\sqrt{77}$

34 [Intermediate] • Target: Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14)

A curve passes through the point $\left(1, \frac{1}{2}\right)$ and satisfies the differential equation $\frac{dy}{dx} = \frac{y^2(1 - \ln x)}{x^2}$ for $x > 0$. Determine the exact coordinates of the stationary point of the curve and its nature.

a. $\left(e, \frac{e}{2e-1}\right)$, which is a local maximum

b. $\left(e, \frac{1}{2e-1}\right)$, which is a local maximum

c. $\left(e, \frac{e}{2e+1}\right)$, which is a local minimum

d. $\left(\frac{1}{e}, \frac{1}{2e+1}\right)$, which is a local minimum

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35 [Intermediate] • Target: Free-Response Questions (FRQs): Challenging & Cross-Lesson Synthesis (15–20)

A curve satisfies the differential equation $\frac{dy}{dx} = \frac{k-y}{2x+1}$ for $x > -\frac{1}{2}$, where k is a real constant. The curve passes through the points $(0, 8)$ and $(4, 2)$. Find the value of k.

a. -1

b. 11

c. $\frac{5}{4}$

d. 5

36 [Intermediate] • Target: Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)

Find the particular solution of the differential equation $\frac{dy}{dx} = 2xe^{-y}$ given that $y = 0$ when $x = 0$.

a. $y = \ln(x^2 + 1)$

b. $y = \ln(x^2)$

c. $y = \ln(2x^2 + 1)$

d. $y = x^2$

37 [Intermediate] • Target: Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37)

Two lines have slopes $m_1 = -1$, $m_2 = -1$ and y-intercepts $b_1 = -2$, $b_2 = -2$. Classify the system.

a. Consistent and independent (one solution)

b. Consistent and dependent (infinitely many solutions)

c. Inconsistent (no solution)

d. Cannot be determined

Part VI — Multiple-Choice Questions (MCQs): Challenging & Cross-Lesson Synthesis (38–45)

38 [Challenging] • Target: Multiple-Choice Questions (MCQs): Challenging & Cross-Lesson Synthesis (38–45) • Cross-Lesson: 19.01 Normal Distribution

Find the projection of $\mathbf{u} = \langle 3, 2 \rangle$ onto $\mathbf{v} = \langle 2, 2 \rangle$.

a. $\left\langle \frac{3}{2}, \frac{5}{2} \right\rangle$

b. $\left\langle \frac{5}{2}, \frac{5}{2} \right\rangle$

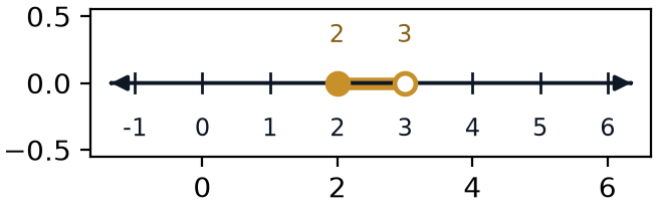
c. $\left\langle \frac{7}{2}, \frac{5}{2} \right\rangle$

d. $\frac{5}{4}$

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39 [Challenging] • Target: Free-Response Questions (FRQs): Foundational Fluency (1–7) • Cross-Lesson: 3.01 Graphs & Transformations

Solve the simultaneous system: $\log_2(x) - \log_2(y) = 2$ and $x - 3y = 3$. What is the value of x?



a. 12

b. 3

c. 6

d. 14

40 [Challenging] • Target: Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14) • Cross-Lesson: 4.01 Coordinate Geometry

Find the angle between $\mathbf{u} = \langle 2, 1 \rangle$ and $\mathbf{v} = \langle 4, 2 \rangle$ (to one decimal place).

a. 0.0°

b. 20.0°

c. 180.0°

d. 10.0°

41 [Challenging] • Target: Free-Response Questions (FRQs): Challenging & Cross-Lesson Synthesis (15–20) • Cross-Lesson: 5.01 Trigonometry

Find the unit vector in the direction of $\mathbf{v} = \langle -8, 15 \rangle$.

a. $\left\langle \frac{9}{17}, \frac{15}{17} \right\rangle$

b. $\left\langle \frac{15}{17}, -\frac{8}{17} \right\rangle$

c. $\left\langle -\frac{8}{17}, \frac{15}{17} \right\rangle$

d. $\langle -8, 15 \rangle$

42 [Challenging] • Target: Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28) • Cross-Lesson: 6.01 Differentiation & Integration

Which differential equation correctly models the statement: "The rate of decrease of the temperature T of an object is proportional to the difference between T and the surrounding temperature T_e."?

a. $\frac{dP}{dt} = kP$

b. $\frac{dT}{dt} = -k(T - T_e)$

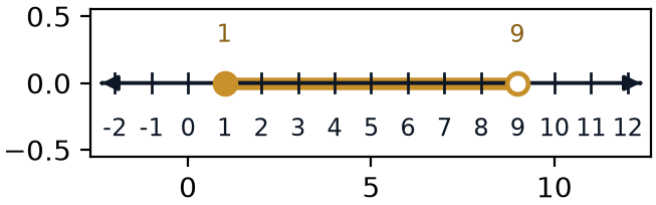
c. $\frac{dN}{dt} = kN(S - N)$

d. $\frac{dm}{dt} = -km$

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43 [Challenging] • Target: Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37) • Cross-Lesson: 7.01 Statistics & Probability

For the hyperbola $\frac{x^2}{4} - \frac{y^2}{9} = 1$, what are the equations of the asymptotes?



a. $y = \pm\frac{3}{2}x$

b. $y = \pm\frac{2}{3}x$

c. $y = \pm 6x$

d. $y = \pm\frac{5}{2}x$

44 [Challenging] • Target: Multiple-Choice Questions (MCQs): Challenging & Cross-Lesson Synthesis (38–45) • Cross-Lesson: 8.01 Functions: Composite & Inverse

Which differential equation correctly models the statement: "The rate of increase of the number of infected people N is proportional to the product of N and the number of remaining susceptible people $(S - N)$, for fixed population S."?

a. $\frac{dm}{dt} = -km$

b. $\frac{dV}{dt} = kr^2$

c. $\frac{dT}{dt} = -k(T - T_e)$

d. $\frac{dN}{dt} = kN(S - N)$

45 [Challenging] • Target: Free-Response Questions (FRQs): Foundational Fluency (1–7) • Cross-Lesson: 9.01 Circular Measure

Which differential equation correctly models the statement: "The rate of increase of the volume V of a balloon is proportional to the square of its radius r."?

a. $\frac{dT}{dt} = -k(T - T_e)$

b. $\frac{dm}{dt} = -km$

c. $\frac{dV}{dt} = kr^2$

d. $\frac{dP}{dt} = kP$

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